

Aug 20, 2025

Healthcare

GLSI

NCM

Rating

Outperform

Initiation

Current Price

\$11.47

Target Price

\$45.00

Market Capitalization

156.34m

Shares Outstanding

13.63m

Float

6.50

Institutional Holdings

9.67%

12-Month Low/High

\$8.06/\$16.50

Average 90-Day Volume

59340

Fiscal Year End

2025-12-31
Revenues (\$ MIL)

Period	2024A	2025E	2026E
Q1	0.0	0.0A	0.0E
Q2	0.0	0.0A	0.0E
Q3	0.0	0.0E	0.0E
Q4	0.0	0.0E	0.0E
	0.0	0.0E	0.0E

EPS (\$)

Period	2024A	2025E	2026E
Q1	(0.19)	(0.25)A	(0.35)E
Q2	(0.20)	(0.30)A	(0.36)E
Q3	(0.20)	(0.36)E	(0.37)E
Q4	(0.60)	(0.34)E	(0.40)E
	(1.21)	(1.30)E	(01.48)E

Greenwich LifeSciences, Inc.

Initiating Coverage With An Outperform Rating and \$45 Price Target

Greenwich LifeSciences Is Developing An Immunotherapy For Prevention Of Breast Cancer Recurrence. Greenwich LifeSciences is a biotechnology company developing GLSI-100, an immunotherapy based on HER2/neu. GLSI-100 completed four clinical trials that lead to the design of the current Phase 3 Flamingo-01 trial. The trial is currently enrolling patients in the US and Europe.

GLSI-100 Is Directed At A Validated Target. GLSI-100 contains GP2, a segment of the HER2/neu (HER2, or human epidermal growth factor receptor 2) receptor found on the surface of breast cancer cells. HER2 is overexpressed in several common cancers, with an estimated 75% of all breast cancers expressing HER2 at some level. Monoclonal antibodies targeting HER2 are the current standard of care for treating certain types of breast cancer.

Phase 3 Flamino-01 Trial Is Ongoing. GLSI-100 is currently being tested in the Phase 3 Flamingo-01 trial for prevention of breast cancer recurrence in high-risk patients. It is given after completion of standard therapy to stimulate proliferation of immune cells that can detect HER2 on cancer cells. If cancer cells start to recur or metastasize, these immune cells can recognize the HER2-positive cancer cells, start a response, and kill them.

We See Large Market Potential For GLSI-100. There are approximately 700,000 new cases of breast cancer each year in the US and EU. The 5-year survival rate is about 80% to 90%, although about 42,000 patients die annually in the US. A vaccine that prevents recurrence could prevent disease progression and death, making it a valuable therapy. Recurrence also leads to retreatment and high costs for the health care system. The reduction in relapse rates could also save large sums for the payors.

Valuation and Conclusion. We estimate market entry by FY2028, and value the stock based on FY2029 EPS. We believe this to be a conservative estimate for the first full year of sales, which could come earlier. Applying a multiple of 15X to the discounted EPS for a target of \$45 per share.

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**Refer to the last two pages for
Analyst Certification & Disclosures**

Investment Thesis

Company Summary. Greenwich LifeSciences is a biotechnology company developing an immunotherapy to prevent recurrence of breast cancer. GLSI-100, the company's lead product, completed four clinical trials before starting the current Phase 3 Flamingo-01 trial.

GLSI-100 contains GP2, a segment of the HER2/neu (HER2, or human epidermal growth factor receptor 2) receptor found on the surface of breast cancer cells. HER2 is overexpressed in several common cancers, with an estimated 75% of all breast cancers expressing HER2 at low (1+), intermediate (2+), and high (3+, HER2 positive) overexpression levels. GLSI-100 delivers GP2 to stimulate an immune response which can detect HER2-positive cancer cells and kill them.

In December 2020, the company announced results of a Phase 2b study showing a 5-year survival rate of 100% (recurrence rate of 0%) in high-risk patients. The Flamingo-01 trial followed this design and is in progress to show GLSI-100 efficacy. Its primary endpoint is prevention of breast cancer recurrence in high-risk cancer patients following surgery.

Phase 3 Flamingo-01 Trial Is Ongoing. GLSI-100 is currently being tested in the Phase 3 Flamingo-01 trial for prevention of breast cancer recurrence in patients with residual disease or at high-risk of recurrence. It delivers the antigen GP2, a 9-amino acid segment of the transmembrane peptide HER2. The immune cell stimulant Leukine (GM-CSF, granulocyte macrophage colony stimulating factor) is given with GLSI-100 to stimulate proliferation of immune cells that detect the GP2 antigen. If cancer cells start to recur or metastasize, these immune cells can recognize the HER2 positive cancer cells, start a response, and kill them.

Patients enrolled in the trial are screened for their level of HER2 expression and by immune criteria (HLA types). Those with the most common HLA type are randomized into either the active or placebo arm. Patients with other HLA types are entered into an open-label arm. Each arm has a target enrollment of about 250 patients, for a total of roughly 750 patients.

Following surgery, patients complete a standard one-year course of Herceptin (trastuzumab, an anti-HER2 antibody) as both neoadjuvant and postoperative adjuvant-based treatment. GLSI-100 is given once per month for six months followed by 5 booster doses every 6 months for a total of 11 doses over 3 years. The first six month period is referred to as the Primary Immunization Series (PIS), in which the immune system starts the process of recognition for its primary immune response. The PIS can be measured by a local skin test and immunological assay. The primary endpoint of the trial is recurrence-free survival.

Flamingo-01 trial began in early 2023. About 40 clinical sites in the US and 80 clinical sites in the EU are enrolling patients. Additional sites are expected to bring the total to 150 actively enrolling sites by the end of 2025. An interim analysis is planned when the reported events reach half of the full number of expected events in the trial.

The Phase 3 Trial Builds On Previous Data. In December 2020, top-line data from the Phase 2b trial showing strong survival was presented at the San Antonio Breast Cancer Symposium. The final efficacy analysis showed patients treated with GP2 immunotherapy following surgery and Herceptin treatment had 100% survival (0% breast cancer recurrences, $p = 0.0338$) over a median of 5 years of follow-up. The Phase 3 Flamingo-01 trial follows this design with increased statistical power. A final report is being prepared for inclusion in the BLA filing, along with additional efficacy and safety data.

Current Drugs Have Improved Outcomes, But Breast Cancer Survival Remains A Problem. Nearly 1 in 8 women in the US will develop breast cancer over her lifetime. There are approximately 700,000 new cases and an estimated total of 9.5 million breast cancer survivors in the US and EU. The 5-year survival rate is about 80% to 90%, with about 42,000 patient deaths annually in the US.

Out of the estimated 700,000 patients diagnosed with breast cancer each year in the US, patients are divided into groups by their different HER2 surface markers and immune characteristics that correlate with immune responses. An estimated 40,000 to 80,000 have the HER2 and HLA characteristics that are within the current Phase 3 Flamingo-01 trial entry criteria. Expansion into additional groups could double the number of patients eligible for therapy.

Market Potential For GLSI-100. A vaccine that prevents local or metastatic breast cancer recurrence could prevent disease progression and patient death, making it a valuable therapy. Breast cancer recurrence leads to hospitalization, retreatment, and high costs for the health care system, so that reducing relapse rates could save large sums for the payors. This would justify high pricing consistent with other immunotherapies.

Valuation and Conclusion. We estimate market entry by FY2028, and value the stock based on FY2029 EPS of \$9.90 per share. We believe this to be a conservative estimate for the first full year of sales, which could come earlier. Applying a multiple of 15X to the discounted EPS for a target of \$45 per share.

Company Background

Greenwich Lifesciences is developing GLSI-100, an immunotherapy to prevent recurrence of breast cancer. Four clinical trials have been conducted, with the Phase III Flamingo-01 trial is in progress for preventing breast cancer recurrence following surgery. GLSI-100 delivers a nine-amino acid sequence of the HER2/neu receptor (HER2, or human epidermal growth factor receptor 2) known as GP2. It is administered with GM-CSF to stimulate proliferation of immune cells against GP-2.

Phase 2b results were presented at the San Antonio Breast Cancer Symposium in December 2020. Following surgery, patients received standard of care therapy with Herceptin for one-year, followed by six months of monthly injections with GLSI-100 and boosters every six months. The final efficacy analysis showed treatment with GP2 immunotherapy had 100% survival (0% recurrences, $p = 0.0338$) over the median 5 years of follow-up.

The Phase 3 Flamingo-01 is a double-blind placebo-controlled trial that began in early 2023. It follows the design of the Phase 2b study, testing GLSI-100 for prevention of recurrence in high risk HER2 positive breast cancer patients. These patients have had residual disease or high-risk pathologic complete response at surgery. Patients enter the trial after surgery and completion of a standard of care treatment with an anti-HER2 antibody as neoadjuvant and postoperative adjuvant treatment.

Out of the estimated 700,000 patients diagnosed with breast cancer each year in the US, patients are divided into groups by their different HER2 surface markers and immune characteristics that correlate with immune responses. An estimated 40,000 to 80,000 have the characteristics that are within the current Phase 3 Flamingo-01 trial entry criteria. Expansion into additional groups could double the number of patients eligible for therapy.

Entry Criteria For Flamingo-01. Tumor cells are tested for expression levels of HER2 surface markers and HLA (human leukocyte antigens) serotype group. HLA proteins are the immune system uses to distinguish between healthy tissue and foreign substances. This testing is intended to ensure that patients have the HLA type needed for a potent immune response, as different HLA types can have variable responses to the HER2 antigen.

Those with the most common HLA type, known as HLA-A*02, are randomized into either the active or placebo arm. Patients with other HLA types are entered into an open-label arm known as "non-HLA-A*02". Each arm has a target enrollment of about 250 patients, for a total of roughly 750 patients. The primary endpoint is recurrence-free survival.

At the current time, about 40 clinical sites in the US and 80 in the EU are enrolling patients. Additional sites have been approved and are opening. The total number of actively enrolling sites is expected to reach 150 by the end of 2025. Target enrollment in each arm is about 250 patients, for a total of 750 patients.

Prospective Interim Analysis Is Based On Event Occurrence. The trial has been designed to detect a hazard ratio (likelihood of the event occurring) of 0.3 in invasive breast cancer-free survival, which will require 28 events to be reported. An interim analysis for superiority or futility will be conducted when at least 14, half of those events, have occurred. This sample size provides 80% power if the annual rate of recurrence events in placebo-treated subjects is 2.4% or greater. This is a conservation assumption, compared with the estimated rate of recurrence, which is about 3.0% to 3.5% per year, or about 11% over the course of the study.

The trial is led by Baylor College of Medicine and currently includes US clinical sites from university-based hospitals and cooperative networks. GP2 was developed by the Henry M. Jackson Foundation For the Advancement of Military Medicine (HJF), a non-profit research institution in Bethesda, Maryland. The company licensed GP2 from HJF in April 2009 as the Phase 2b trial was in progress.

Greenwich Lifesciences completed its IPO in September 2020, raising about \$36.5 million from the IPO and its December 2020 follow-on offering. This has kept the number of shares outstanding relatively low. Management, directors, and certain other shareholders own 52% of the company, with a lock-up and leak-out agreement that started with the December 2020 IPO and has been extended through April 2026. In addition, an additional 20-30% of the shares outstanding are owned by friends and family.

Background on Breast Cancer and HER2/neu

Breast Cancer Market Size and Incidence. An estimated 700,000 new cases of breast cancer are diagnosed each year, with about 9.5 million breast cancer survivors in the US and EU. Upon diagnosis, tumors are tested to determine the genetic origin and characteristics of the cancer cells. One of these tests classifies the tumor by the type of surface receptors that feed tumor cell growth.

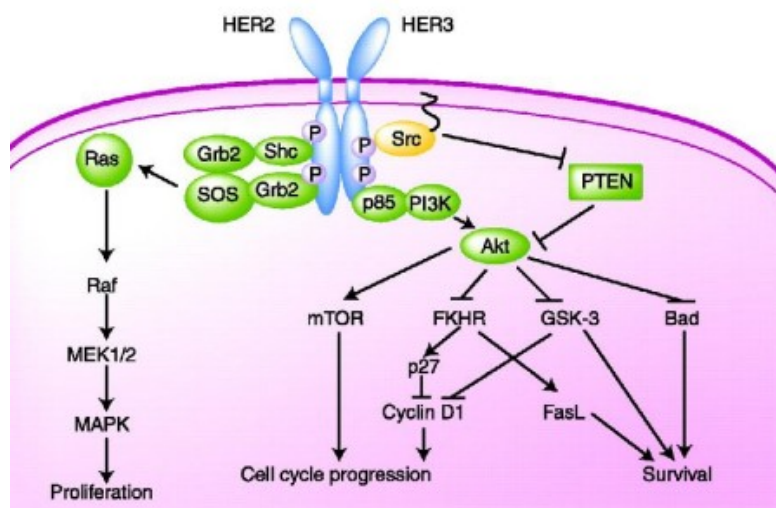
There are four surface receptors present on the surface of the cell that can be overexpressed and lead to tumor growth. These are the estrogen receptor (ER), progesterone receptor (PR) or human epidermal growth factor receptor 2 (HER2/*neu*). Tumors with ER and/or PR overexpression are known as hormone receptor (HR+) positive tumors. These can be treated with drugs (tamoxifen and letrozole for ER) that block the hormones from binding the receptor, stopping the growth signal. About 25% to 30% of all breast cancers have HER2 overexpression, with higher overexpression levels correlating with more aggressive tumors and poor prognosis. These can be treated with monoclonal antibodies developed against the HER2/*neu* receptor. An estimated 15% to 20% lack all three receptors and are known as “triple negative or TNBC” tumors, which are the most aggressive and difficult to treat.

HER2/*neu* is a receptor on the cell surface that normally functions to promote cell growth and division. It is a member of the epidermal growth factor (EGFr) receptor family that binds epidermal growth factor signaling molecules. Under normal conditions, the receptor binds its target and undergoes several steps that activate biochemical pathways. These intracellular pathways carry the signals to the nucleus, where they turn genes on and off to initiate cell functions, such as cell growth and proliferation.

Product Description & Mechanism of Action

HER2/Neu Is A Validated Target In Breast Cancer. The surface protein HER2/*neu* is overexpressed in a variety of common cancers. An estimated 75% of all breast cancers express HER2 at low (1+), intermediate (2+), and high (3+, HER2 positive) overexpressor levels. HER2 receptor overexpression is also found in ovarian, stomach, some forms of uterine cancer and lung cancers, and in 30% of salivary duct carcinomas, but with lower overexpression levels or frequencies.

Exhibit 1. HER2/neu Signals Gene Activation. One of the first steps is changes in molecular conformation and binding to an adjacent HER2 receptor, forming a dimer. This leads to phosphorylation of the receptor and starts several signaling pathways that lead to proliferation and growth of the cancer cells. These pathways include the MAP kinase and the PI3 kinase/AKT pathways, which activate the NF- κ B pathway. Signals on these pathways promote cell proliferation and the growth of blood vessels to nourish the tumor (angiogenesis). HER2 overexpression can also overcome regulatory checkpoints in normal cells mitosis (division) that prevent overproliferation, known as CDK2 proteins.



Source: Greenwich LifeSciences, Inc.

When HER2 is overexpressed, it can reach 100 times its normal levels. HER2-positive breast cancers have poor prognosis and are associated with increased disease recurrence. Mutations in the receptor can cause the pathways to be overactive, leading to unregulated growth and progression. There are several drugs used in breast cancer therapy that target HER2:

- **Herceptin** (trastuzumab, from Genentech/Roche, approved September 1998) is a monoclonal antibody against the HER2/*neu* receptor on cancer cells. It binds to the receptor, stopping the intracellular pathway that signals replication in the cell nucleus. The current standard of care for patients that have HER2 positive breast cancer is for surgery followed by monthly injections of Herceptin for the first year. This adjuvant (after primary treatment) treatment has been shown to reduce recurrence rates from 25% to 12% (50% improvement). In the neoadjuvant setting (before surgery, induction therapy), HER2/*neu* positive patients are given treatment before surgery. A surgical biopsy is taken, and after surgery the patient may receive either Herceptin or Kadcyla, a HER2 antibody linked to a cytotoxic drug.
- **Perjeta** (pertuzumab, from Genentech/Roche, approved June 2012) is a monoclonal antibody similar to Herceptin that binds to several different members of the HER receptor family to block heterodimers (HER2/HER3) from forming. In contrast, Herceptin only binds to HER2 and blocks homodimers (HER2/HER2) from forming. Perjeta is used in combination with Herceptin and docetaxel for the treatment of metastatic HER2-positive breast cancer. It is also used in the same combination as a neoadjuvant in early HER2-positive breast cancer.
- **Kadcyla** (emtansine, from Genentech/Roche, approved February 2013) is a Herceptin derivative containing the trastuzumab antibody linked to DM1, a cytotoxic drug. The trastuzumab portion binds to the HER2 receptor to block receptor dimerization and stop cell signaling. The molecule is internalized and broken down inside the cell. This releases DM1, which binds the tubulin proteins required for mitosis. This stops the mitotic process and kills the cell. Since HER2 is over-expressed in cancer cells, the DM1 is only taken up by tumor cells. Kadcyla has been shown to reduce recurrence rates from 22% to 11% (50%) in the neoadjuvant setting.

While Herceptin adjuvant therapy after surgery can reduce recurrence rates from 25% to 12%, about 50% to 70% of HER2+ patients do not respond to treatment. Virtually all patients develop resistance that reduces efficacy over time. The 12-month treatment requires office visits for a 90-minute infusion, and commonly leads to fatigue, joint pain, and diarrhea. Some have infusion reactions, including flu-like fever and chills. Cardiac side effects are less common but can be serious.

GLSI-100 Was Designed To Recognize Cells Bearing The HER2 Surface Marker

GLSI-100 contains GP2, a nine-amino acid transmembrane portion of the HER2/*neu* protein. Following first-year treatment with Herceptin (or other anti-HER2 treatments), GLSI-100 is administered as an intradermal injection along with Leukine (GM-CSF granulocyte macrophage colony stimulating factor) to stimulate an immune response. This combination leads to proliferation of the cells required for antigen recognition, immune cell activation, and immune memory. As the immune cells circulate throughout the body, any cells with the GP2 antigen can be identified as recurrent cancer cells, starting an immune response to kill them.

The GLSI-100 Mechanism of Action. The combination of GP2 and GM-CSF stimulates the primary immune response against the GP2 antigen. Upon GLSI-100 administration to a patient, the GP2 antigen is recognized by antigen presenting cells (APCs). The APCs engulf the antigen and internalize it.

Once the APC internalizes the antigen, breaks it into pieces, and adds molecules known as major histocompatibility complexes (MHCs). The MHCs bind to the GP2 antigen and are then transported to the cell surface. Next, the GP2 antigen-HLA complex is displayed on the cell surface. When inactive CD8+ T cells that circulate through the body recognize the antigen complex, they are activated and mature into CD8+ T cells. This creates a population of T cells that can recognize cells with GP2 on their surface as HER2-positive cancer cells. As these T cells circulate throughout the body, they identify and destroy cells bearing the GP2 antigen, preventing the recurrence of HER2 positive cancer.

GM-CSF also enhances monocyte and neutrophil cytotoxicity against the tumor cells and magnifies activity-dependent cellular cytotoxicity (ADCC) of monocytes and neutrophils against targets coated with the anti-ganglioside antibodies. Preclinical studies have shown that T cells sensitized against the GP2 peptide demonstrate significant recognition of HER2-expressing tumors. Both ovarian and breast cancer-specific cytotoxic lymphocytes (CTLs) recognize GP2, which is widely expressed in HER2-expressing tumors and is capable of inducing tumor-specific CTL populations in vitro.

Exhibit 2. GP2 Peptide Is A Subunit of HER2/*neu*. GP2 is a 9 amino acid HER2/*neu* transmembrane peptide that elicits a targeted immune response against HER2/*neu*-expressing cancers. The left panel shows an image of a cell surface with several relevant cell surface proteins. Breast cancers and other solid tumors with elevated expression of HER2/*neu* protein are highly aggressive with an increased disease recurrence and poor prognosis. The panel on the right shows the nine amino acid sequence known as GP2, a segment of the HER2 protein.

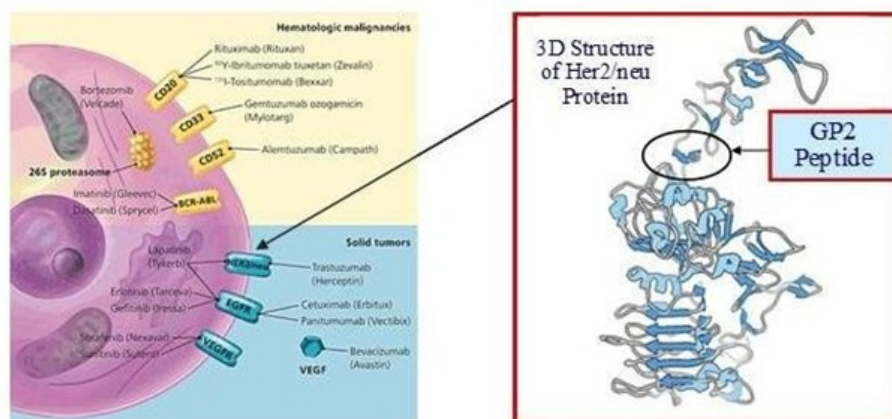
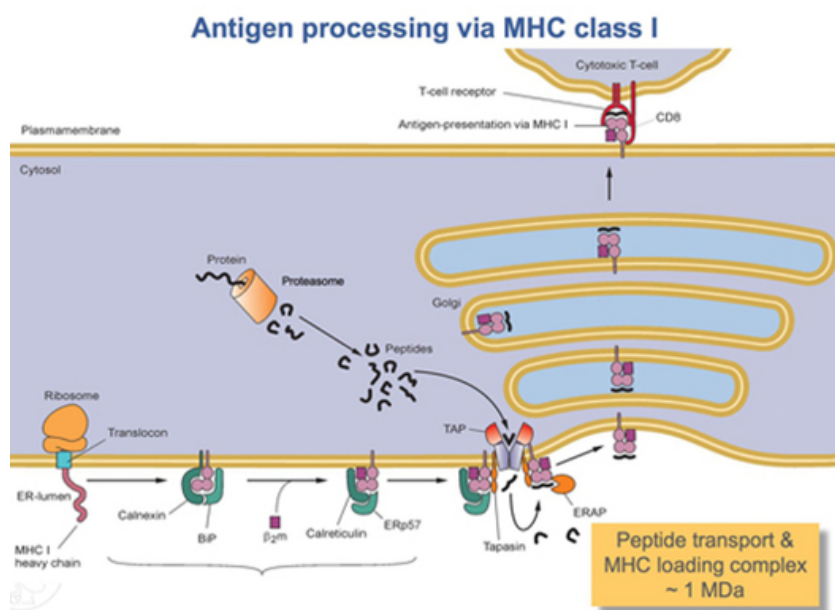


Exhibit 3. Antigen Presentation In The Immune Response. An antigen presenting cell (APC) internalizes an antigen, breaks it into smaller fragments, and adds the MHC molecules to form a complex that is presented on the cell surface. The inactive or immature T cell detects the antigen-MHC complex and develops into a CD8+ killer cell that can recognize and destroy cells bearing the antigen.



Source: Robert Tampé - Molecular Understanding of Transmembrane Processes

The Flamingo-01 Trial Separates Patients By Human Leukocyte Antigen Variations

The MHC system in humans is also known as the Human Leukocyte Antigen, or HLA system. Patient screening in the Flamingo-01 trial includes categorization by HLA type, an immune classification to ensure the patient has a potent immune response to the BP2 antigen. These are proteins that the immune system uses to distinguish between normal cells and those that are foreign or abnormal (known as “self from non-self”). The immune system has 12 unique HLA markers (proteins) that can create millions of possible combinations.

MHC is a group of genes that encode cell surface proteins that process antigens and display them on the cell surface. As described above, these complexes are presented to other immune cells, leading to their activation. There are two main classes of MHC molecules, known as Class I and Class II.

- The Class I molecules recognize proteins from inside the cell, including those that allow the immune system to distinguish normal tissue (“self”) from viruses and bacteria. With Type I, the major groups are known as HLA-A, HLA-B, and HLA-C.
- MHC Class II molecules have similar activity but mostly process extracellular proteins that rather than those from inside the cell. The major types of Class II are HLA-DR, HLA-DQ, and HLA-DP.

The MHC genes have many variations that express different subtypes of MHCs among the population. In the HLA-A group, over 2,400 alleles have been identified. This allows for an infinite number of combinations for recognizing antigens. The alleles have been numbered to identify their serotype groups, such as HLA-A*02. These different HLA subtypes lead to responses to cancer between different patients.

The most common HLA type is HLA-A*02, which has been observed in about 46% of the patients enrolled in the Flamingo-01 Phase 3 trial. The frequency of being “double HLA-A*02” (receiving the HLA-A*02 from each parent) is about 8%. The next most common HLA types are HLA-A*03, HLA-A*24, and HLA-A*01, with about 20% to 25% frequency for each group. Frequency of HLA-A*11, HLA-A*68, HLA-A*29, and HLA-A*30 is about 9 to 12% each.

The Flamingo-01 Trial Has Patient Arms By HLA Type. Upon enrollment in the trial, patients are screened for their HER2/neu overexpression level and their HLA serotype. Patients that are in the HLA-A*02 category are randomized to receive either drug or placebo. Patients with other HLA types are entered into an open-label arm known as “non-HLA-A*02”.

Patients enrolled that are HLA-A*02 and who self-report their race, show HLA-A*02 prevalence in white or Hispanic patients of 50%, prevalence in African American patients of 40%, and prevalence in Asian patients of 17%.

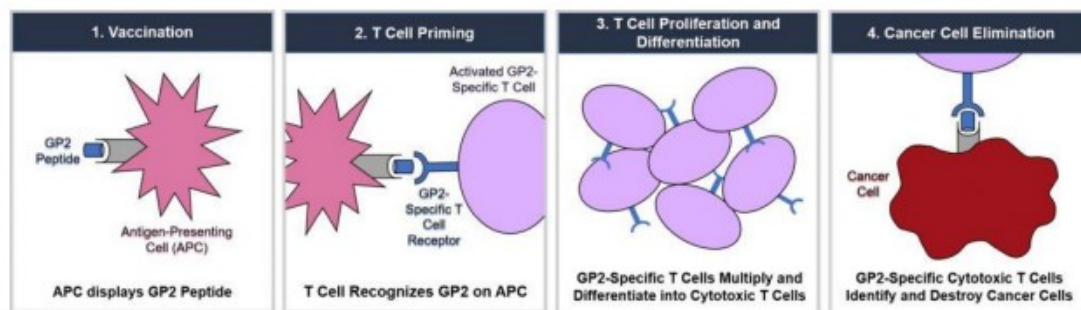
Exhibit 4. Distribution of the HLA Serotypes. The table below shows the frequency of Class I MHC alleles HLA in the North America.

Frequency of Class I MHC alleles HLA in the North America - Data from HLA Matchmaker

Allele	White	Allele	Black	Allele	Hispanic	Allele	Asian/ Pacific
A*02:01	45.6%	C*04:01	29.0%	A*02:01	37.1%	A*11:01	38.4%
C*07:01	27.7%	C*07:01	25.4%	C*04:01	25.4%	A*24:02	33.7%
A*01:01	27.4%	C*06:02	23.0%	A*24:02	24.9%	C*07:02	33.3%
A*03:01	23.8%	A*02:01	22.3%	C*07:02	24.2%	C*01:02	27.7%
C*07:02	21.5%	A*23:01	20.7%	C*07:01	20.8%	A*33:03	23.3%
C*04:01	21.2%	C*02:02	19.0%	C*03:04	14.4%	C*08:01	21.6%
B*44:02	20.2%	A*03:01	18.7%	A*03:01	14.3%	C*03:04	19.9%
B*07:02	18.1%	C*07:02	18.1%	B*07:02	13.2%	A*02:01	18.1%

Source: HLA Manchester, Greenwich LifeSciences, Inc.

Exhibit 5. GLSI-100 Mechanism of Action. GP2 is administered in combination with the FDA-approved immunoadjuvant GM-CSF to stimulate proliferation of APCs (antigen presenting cells). Following GP2 administration, a population of CD8+ cytotoxic T lymphocytes develops that recognizes and destroys HER2/neu-expressing cancer cells. Preclinical studies have shown that T cells sensitized against the GP2 peptide demonstrate significant recognition of HER2/neu-expressing tumors.



Source: Greenwich LifeSciences, Inc.

Clinical Development and History. The GP2 technology was licensed by GLSI from the Henry M. Jackson Foundation for the Advancement of Military Medicine (HJF), a non-profit organization created by Congress in 1983. HJF conducts research, manages clinical trials, and out-licenses technology developed by the US military. This is intended to connect the military medical community with private partners to develop and commercialize the foundation's research.

HJF and Harvard University developed the GP2 technology and conducted three Phase 1 and one Phase 2b clinical trials at Walter Reed Army Medical Center, MD Anderson Cancer Center, and various other military and civilian sites. HJF managed regulatory filings, trial data, and manufacturing. Greenwich LifeSciences in-licensed GP2 in 2009 as the Phase 2b trial was in progress. It became the regulatory sponsor, while responsibility for conducting the trial and for the data from the Phase IIb trial remains with HJF.

In the licensing agreement, GLSI received all of the GP2 data from the Phase IIb trial and prior trials. It holds exclusive worldwide rights to the patent estate, covering methods of using GP2 as an immunotherapy that elicits a targeted immune response against HER2/*neu*-expressing cancers. HJF received 202,619 shares of GLSI common stock, with an annual maintenance fee and milestone payments of up to an aggregate of \$5.7 million plus a royalty on sales of 2.5% to 5% depending on volume, and reimbursement for patent expenses. The license term runs until the expiration of the patents. No milestone or royalty payments have been made yet.

Phase 2b Trial Data Presentation

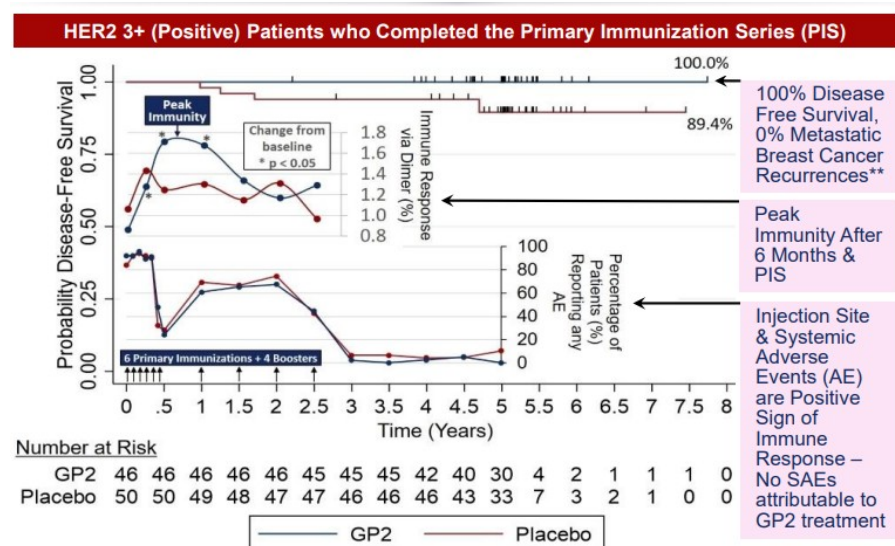
In December 2020, the company presented 5-year survival data from the Phase 2b trial at the San Antonio Breast Cancer Symposium. The poster included data from 168 patients at 16 sites and met all of its clinical endpoints.

Study Design and Treatment. The study enrolled 180 patients across 16 clinical sites with both high (HER2 3+) and low HER2 expression (1-2+). Patients had surgical resection followed a standard 1-year course of Herceptin. GLSI-100 was given as a Primary Immunization Series (PIS) of 6 intradermal injections to demonstrate safety and elicit an immune response, with a booster every six months for a total of 11 treatments over three years.

After 5 years of follow-up in patients that completed the PIS, the Kaplan-Meier estimated 5-year disease-free survival (DFS) rate in the 46 HER2/*neu* 3+ patients treated with GLSI-100 was 100%. This compares with 89.4% (95% CI:76.2, 95.5%) in the 50 placebo patients treated with only GM-CSF ($p = 0.0338$).

Safety. GLSI-100 was shown to be well tolerated with no serious adverse events related to study medication. Patients had potent responses to DTH (Delayed-Type Hypersensitivity) tests, a method of testing for cell-mediated immunity to a specific antigen. Redness or inflammation in the test area is an indication that the immune system had been sensitized to the antigen and contained memory T-cells that could react upon future exposure. As expected, injection site reactions were common, occurring in almost 100% of patients treated with either GLSI-100 or GM-CSF alone. The study showed that GLSI-100 was able to stimulate a potent immune response from the PIS. This correlated with disease-free survival.

Exhibit 6. Five Year Data Set of GP2 Phase 2b Trial Showed 100% Survival With No Recurrence



The Phase 3 Flamingo-01 Trial Is In Progress

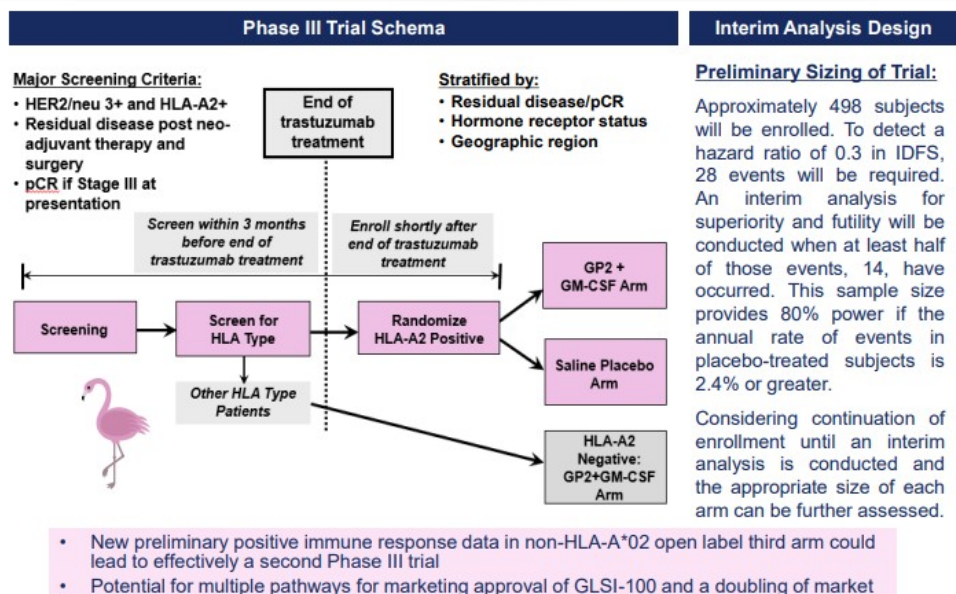
The data from the Phase 2b trial was then used to design a large Phase 3 trial to support a BLA filing. The trial, known as the Flamingo-01 trial (NCT05232916), is a double-blinded, placebo-controlled study testing GLSI-100 in patients who are at high risk of breast cancer recurrence (residual disease or high risk pathologic complete response at surgery).

- Patients are screened for their HER2 expression level and HLA type. Those that are HER2/neu 3+ and HLA-A*02 positive are randomized into two arms, receiving either GLSI-100 with GM-CSF or the control arm receiving saline placebo.
- Patients that have an HLA type other than HLA-A*02 are entered into an open label arm referred to as “non HLA-A*02”. The non-HLA-A02 types that are most commonly being enrolled in the trial include HLA-A*03, HLA-A*24, HLA-A*01, HLA-A*11, HLA-A*68, HLA-A*29, HLA-A*30, HLA-A*23, and HLA-A*33.

The patients are required to complete standard of care with Herceptin (or other HER2 therapy) for one year in the neoadjuvant and postoperative adjuvant therapy. After completing the year of therapy, patients begin the GLSI-100 stage of treatment. The course of therapy consists of 6 intradermal injections (or PIS, Primary Immunization Series) over the first 6 months. This stimulates the immune response and is followed by 5 booster intradermal injections spaced 6 months apart. The third open-label arm is gathering data to determine if GLSI-100 can be used in non-HLA-A*02 positive patients.

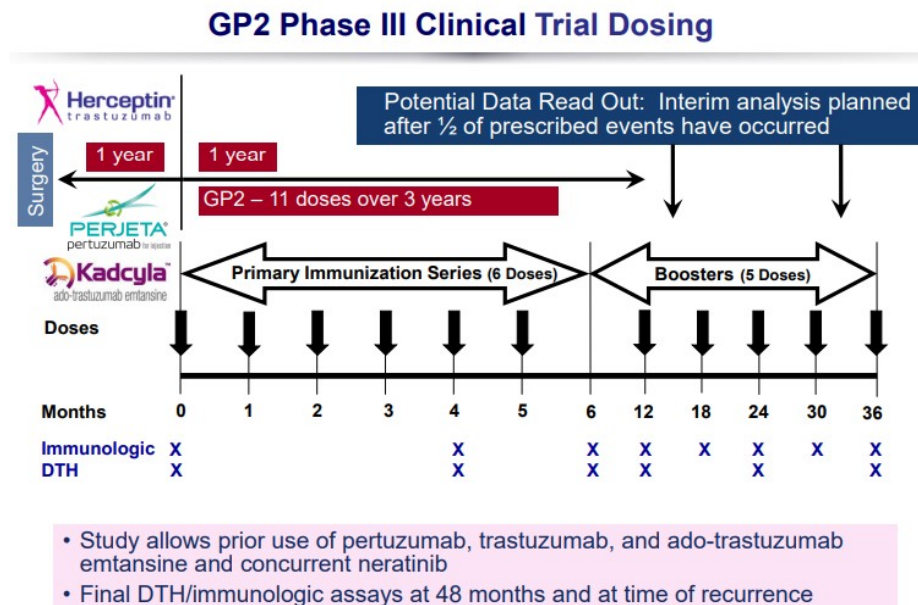
Exhibit 7. The Phase 3 Flamingo Trial Design. The Phase 3 clinical trial was designed to treat a similar population of moderate to severe breast cancer patients as the Phase 2b study.

Flamingo-01 - Phase III Trial Schema & Interim Analysis



Source: Greenwich LifeSciences, Inc.

Exhibit 8. GP2 Phase 3 Clinical Trial Dosing. Patients enrolled in the trial have surgical resection followed by antibody treatment and then the Primary Immunization Series (PIS) for 6 months. This is followed by boosters every 6 months for three years for a total of 11 doses over 36 months of therapy.



Source: Greenwich LifeSciences, Inc.

The trial began in early 2023 and has added clinical sites around the world. There are currently about 40 centers in the US and 80 in the EU actively enrolling patients. Another 30 sites have been approved in the EU, with the total of about 150 sites expected to be enrolling patients by 4Q2025. The trial has a target enrollment of about 498 patients in the placebo-controlled arms and another 250 in the open-label arms.

Exhibit 9. Flamingo-01 Has About 120 Active Clinical Sites Around The World. The trial sites are led by Baylor Medical Center and a consortium of prominent clinical centers.

Participating networks: US Oncology/Sarah Cannon, TRIO-US, GEICAM (Spain), Unicancer (France), GBG (Germany), GIM (Italy)



Source: Greenwich LifeSciences, Inc.

Primary Endpoint/Outcome Measure: The primary endpoint is Invasive Breast Cancer-free Survival (IBCFS). This is defined as time from the first dose of study medication until the date of any recurrence or death. Events that are defined as recurrence include invasive breast cancer recurrence, local-regional invasive breast cancer recurrence, distant recurrence, contralateral invasive breast cancer, or any cause of mortality. The median follow-up is expected to be 4 years with interim analysis planned after a predetermined number of events.

Announcement of Preliminary Flamingo-01 Safety Data. In April 2025, GLSI announced a summary of safety data in patients treated until that point. The study remained blinded with data coming from both arms in the HLA-A*02 group and open-label non-HLA-A*2 group. Interestingly, the immune responses reported for all patients at baseline and the increasing immune response over the Primary Induction Series (PIS) were highly comparable to the data in the Phase IIb clinical trial for HLA-A*02-only patients.

These data show the most frequent adverse event is injection site reaction. This is an expected indication of an immune response and is similar to response seen in the Phase 2b trial. GP2 has been well-tolerated and there have been no safety signals that would be dose-limiting or lead to patients dropping out of the trial.

Patients in all groups have been showing an increase in injection site reaction or DTH (Delayed Type Hypersensitivity) skin test reaction with induration (hardness or immune swelling) or erythema (redness) as more vaccinations are given. As patients received the six monthly vaccinations, an increasing number of patients had immune responses from baseline through the 4th to 6th months in all groups. We believe this indicates GLSI-100 is resulting in an immune response.

Flamingo-01 Phase III Trial Has An Interim Analysis. A prospective interim analysis has been designed into the trial to determine superiority or futility. The trial was designed to detect a hazard ratio (likelihood of an event occurring) of 0.3 in Invasive Breast Cancer-free Survival, which requires 28 events to be reported. The interim analysis will be made when 14 of the 28 events have been reported.

The patient population size provides 80% power if the annual rate of events in placebo-treated subjects is 2.4% or greater. This design 2.4% event rate in the placebo-treated patients is lower than the expected rate from published studies of 3.0% to 3.5% per year, or about 11% over the course of the trial (as seen in Phase 2b). This allows for better-than-expected survival in the placebo group, which we believe is a conservative design.

Upcoming Milestones and Activities

GLSI is currently working to activate clinical sites in Europe that have received regulatory approval to begin patient treatment. As the number of sites grows, the rate of patients recruited each quarter can increase.

The company has not released the number of patients enrolled in the trial, as it is blinded and has both placebo-controlled arms and an open-label arm. Depending on the results seen, the open-label “non-HLA-A*02” arm could be separated to become an arm of a second trial. This could potentially double the number of patients eligible for GLSI-100 therapy. Data from the initial immunology testing and open-label trials could potentially be presented at a medical meeting.

GLSI is also working on activities needed to commercialize GP-100. This includes pre-marketing activities, such as market surveys and KOL discussions. The company does not have a sales force and may begin discussions about a marketing partnership or licensing.

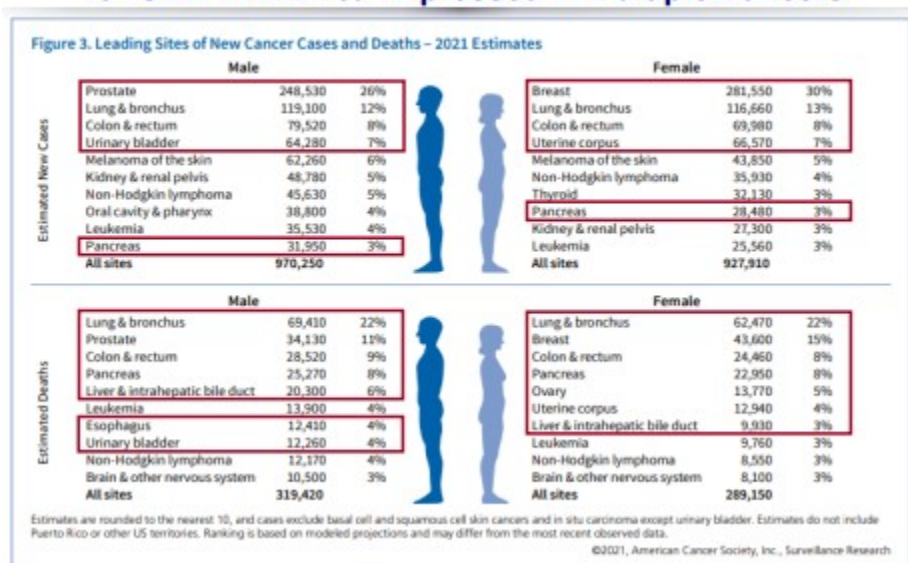
The first large-scale commercial production lot has been completed. The company is also continuing its dialogue with the FDA and European regulators in preparation for a BLA submission.

Pipeline of Indications Could Expand The Market. The HER2 surface market is expressed in many tumor types at variable levels. The Flamingo-01 trial enrolls patients with high overexpression levels (3+), although patients with levels of 1+ and 2+ may also benefit. Expansion into these patient groups would potentially increase the market by 25% to 75%. The HLA-A*02 serotype is about 46% of the population, while other serotypes in the open-label arm could increase the target market to 80% to 100% of the patients.

HER2/*neu* is also in other tumor types, although not as frequently and at variable overexpression levels. GLSI-100 has potential use in these other tumors, especially those that have low response rates and poor prognosis.

Exhibit 10. Additional Potential Indications For GLSI-100. HER2/*neu* is found in many tumor types and has additional potential indications.

Potential Commercial Opportunities / Additional Indications for GP2: HER2/*neu* Expressed in Multiple Cancers



Source: Greenwich LifeSciences, Inc.

Market Potential and Sales Estimates. We have modeled revenues based on the population in the double-blind arms of the Phase 3 trial. Our target population is based on an estimated 700,000 patients diagnosed with breast cancer each year in the US and Europe. We have assumed 25% are HER2 level 3+ overexpressors and 50% are HLA-A*2. This reduces the population to 175,000 patients. Assuming 25% have residual disease or are at high risk gives population of 43,750 within the current trial entry criteria.

We do not include patients in the open-label arm with other HLA types, although these patients could be added. These patients could double the size of the market, but we have excluded these patients.

We have allowed for the Phase 3 trial to continue through FY2026, then allow 12 months for the NDA and 6 months for FDA review. We anticipate revenues in 2028 with market penetration building over the years that follow.

Our models are based on the treatment regimen with 6 monthly doses over two quarters followed by a booster dose at months 12, 18, 24, 30, and 36. We assume patients added evenly over each quarter with an average 2 doses per quarter, assuming patients receive 1, 2, or 3 doses. We expect the number of booster patients to increase after the first year of introduction, with patients added each quarter becoming ready for boosters over the following two and a half years.

- In the first year, revenues are from patients receiving their first course of six monthly doses.
- In year two, patients that started in the first and second quarter that reach 12 months get their first and second boosters in Q1 and Q3 or Q2 and Q4. Patients who received their first doses in the second half of the year receive their 12-month booster in Q3 and Q4. In the quarters of second half of the year, two groups will be receiving their first or second booster.

- In year 3, the patients who began in the first half of previous year receive their first booster in Q1 or Q2, as other patients receive boosters three in the first half of year 1 and four in the second half, as others receive boosters two and three.
- In the first two quarters, three groups will receive boosters for 24, 18, or 12 months, while in the third and fourth quarters, four groups will receive boosters for 30, 24, 18, or 12 months.
- By the fourth year after introduction, there will be patients reaching all five booster intervals in each quarter.

Valuation and Financial Analysis. We have used a price of \$10,000 per treatment, which we believe is comparable to other cancer therapies. Based on an improvement in relapse-free survival from 90% to 97%, we believe the cost of treating the population compares favorably with costs of treating relapsed patients. We also believe the cost per year of life saved is comparable to other medical treatments.

Our assumptions are used to project the target population and market penetration to estimate the number of patients treated per quarter. We then project revenues, which we discount at 50% to allow for clinical risk.

These revenue projections are then used in our Income Statement. We have applied conservative assumptions for Cost of Goods Sold at 20% in the first year, allowed for ongoing research, with marketing and selling expenses to estimate earnings per share. We then further discount these estimates at 35% per year to allow for financial, clinical, and regulatory risk.

Conclusion. We estimate market entry by FY2028, and value the stock based on FY2029 EPS of \$9.90 per share. We believe this to be a conservative estimate for the first full year of sales, which could come earlier. Applying a multiple of 15X to the discounted EPS for a target of \$45 per share.

Risk Factors

Risks to our rating and price target include, but are not limited to:

Drug development risk: Greenwich LifeSciences is a development stage company conducting clinical trials for its lead product, GLSI-100. The company faces the risks of the drug development industry, including scientific, technical, clinical, regulatory failures. As novel therapies, the drugs also face risks with reimbursement and product adoption.

Company risks: The company has incurred significant losses and negative cash flow operations since inception and expects to incur losses and negative cash flows for at least the next 12 months. Revenues are not expected for several years and it is dependent on its ability to raise additional capital. The company is dependent on the success of its technology platform, including their lead product candidate, GLSI-100. For these reasons, the independent public accounting firm has expressed doubt about its ability to continue as a going concern.

Greenwich LifeSciences is a Smaller Reporting Company and an Emerging Growth Company, as defined in Section 2(a) of the Securities Act, as modified by the JOBS (Jumpstart Our Business Startups) Act of 2012. This may make GLSI's financial statements difficult to compare with other public companies.

International risks: The international aspects of their business expose the company to business, regulatory, political, operational, financial, and economic risks associated with doing business outside of the US. GLSI is conducting clinical trials both inside and outside the US. Unfavorable global economic conditions could adversely affect their business, financial conditions, or results of operations.

Intellectual property risk: The field of patents and intellectual property involves complex scientific and legal issues that are subject to change by Congressional legislation or judicial action. Other companies with greater resources may challenge the company's intellectual property estate through the legal system or through the US Patent and Trademark Office. Outcomes of such challenges are difficult to predict and subject to reversal on appeal. The company holds a patent portfolio and has licensed patents from other parties, on which it depends for its proprietary positions. While the company has no pending intellectual property challenges, this is a risk inherent to the industry.

Clinical supplies and manufacturing risk: Greenwich LifeSciences leases its operating facilities and depends on clinical trial managers and third-party suppliers for its clinical trial grade materials, including the active pharmaceutical ingredients. We believe the supply of GLSI-100 clinical materials is sufficient to conduct the trials, but third-party manufacturing still carries a risk of problems or disagreements that could cause delays.

Regulatory risk: The lead product, GLSI-100, has been through pre-clinical trials and Phase 2b clinical trials. Although we believe the clinical data indicate efficacy, further testing is needed before market approval. Each step in the clinical development process requires interaction with the FDA, which may not agree with the company's development plans.

GLSI is in Phase 3 clinical trials. Data and findings from the clinical trials are submitted as part of an application to the FDA for marketing approval. The application must be reviewed by the FDA before the company receives approval for marketing. Analysis by the FDA may not agree with the analysis presented by the company. Approval of the application cannot be assumed.

Exchange and market risk: GLSI shares trade on the NASDAQ exchange and have relatively small daily volume. The company is expected to raise additional capital to fund operations before its products reach the market, which is subject to market conditions.

Directors, executive officers, and principal stockholders and affiliates beneficially own approximately 52% of the outstanding common stock. These stockholders may be able to control the outcome of matters submitted to stockholders for approval, including the election of directors and any merger, consolidation, or asset sales. These stockholders may be able to influence management and decisions at the company. The views and priorities of these shareholders may differ from the general public.



Legislation and policy changes: Laws for drug approval are established by Congress and administered by the FDA. Reimbursement by third-party payors often follows policies established by the Center for Medicaid/Medicare. Both agencies are divisions of the Department of Health and Human Services, run by Commissioners appointed by the President and confirmed by the Senate. The Department of Health and Human Services, the FDA, and CMS have undergone personnel changes during the past 12 months. Regulations and policies are subject to change so that future policies may differ from past policies and expectations.

Company Profile

Company Description. Greenwich LifeSciences is a biotechnology company developing GLSI-100, an immunotherapy to prevent recurrence of breast cancer. GLSI-100 completed four clinical trials before starting the current Phase 3 Flamingo-01 trial. The trial began in January 2023 and is currently enrolling patients in the US and Europe.

Valuation Summary

Our valuation is based on completion of the Phase 3 Flamingo-01 trial, followed by market entry by FY2028. We value the stock based on FY2029 EPS of \$9.90 per share. We believe this to be a conservative estimate for the first full year of sales, which could come earlier. Applying a multiple of 15X to the discounted EPS for a target of \$45 per share.

Greenwich LifeSciences, Inc. Income Statement (in thousands, except per share data)															
Fiscal Year Ended December 31	2023A	2024A	1Q25A	2Q25A	3Q25E	4Q25E	2025E	1Q26E	2Q26E	3Q26E	4Q26E	2026E	2027E	2028E	2029E
Revenues															
GP2 Sales															
HER2 HLA-A2												8,750	140,000	310,625	
Other															
Total Revenues												8,750	140,000	310,625	
Expenses															
Cost of goods sold															
COGS/Revenues													27,519	51,034	
													20%	16%	
Research and development	7,699	12,952	2,601	3,508	3,900	4,200	14,209	4,400	4,600	4,800	5,300	19,100	24,900	33,000	38,700
General and administrative	1,629	3,060	681	538	925	1,050	3,194	950	950	950	950	3,800	6,700	17,600	19,800
Marketing and selling															
MSS/Revenues													21,000	46,594	
													15%	15%	
Total expenses	9,328	16,012	3,282	4,046	4,825	5,250	17,403	5,350	5,550	5,750	6,250	22,900	31,600	78,119	156,128
Operating Income (Loss)	(9,328)	(16,012)	(3,282)	(4,046)	(4,825)	(5,250)	(17,403)	(5,350)	(5,550)	(5,750)	(6,250)	(22,900)	(22,850)	61,881	154,497
Interest income	436	223	24	21	20	18	83	26	27	22	20	95	66	84	88
Total other income	436	223	24	21	20	18	83	26	27	22	20	95	66	84	88
Pretax Income	(8,892)	(15,789)	(3,258)	(4,025)	(4,805)	(5,232)	(17,321)	(5,324)	(5,523)	(5,728)	(6,230)	(22,805)	(22,784)	61,965	154,585
Net Income	(8,892)	(15,789)	(3,258)	(4,025)	(4,805)	(5,232)	(17,321)	(5,324)	(5,523)	(5,728)	(6,230)	(22,805)	(22,784)	61,965	154,585
GAAP EPS (basic)	(0.69)	(1.21)	(0.25)	(0.30)	(0.36)	(0.34)	(1.25)	(0.35)	(0.36)	(0.37)	(0.40)	(1.48)	(1.47)	3.98	9.90
GAAP EPS (diluted)	(0.69)	(1.21)	(0.25)	(0.30)	(0.36)	(0.34)	(1.25)	(0.35)	(0.36)	(0.37)	(0.40)	(1.48)	(1.47)	3.98	9.90
Weighted Average Shares (basic, in th)	12,848	13,015	13,171	13,371	13,384	15,398	13,831	15,413	15,429	15,444	15,459	15,436	15,498	15,560	15,622
Weighted Average Shares (diluted, in th)	12,848	13,015	13,171	13,371	13,384	15,398	13,831	15,413	15,429	15,444	15,459	15,436	15,498	15,560	15,622

Source: Company SEC filings and Noble Capital Markets estimates

Greenwich Lifesciences, Inc: Balance Sheet (in thousands)															
Assets	2023A	2024A	1Q25A	2Q25A	3Q25E	4Q25E	2025E	1Q26E	2Q26E	3Q26E	4Q26E	2026E	2027E	2028E	2029E
Cash and Cash Equivalents	\$6,989	\$4,092	2,750	3,125	2,731	23,520	\$23,520	\$20,381	\$17,143	\$13,800	\$10,255	\$10,255	(\$2,986)	\$47,526	\$211,660
Acquired patents, net	5	2	1	1	1	1	1								
Total current assets	\$6,995	\$4,094	\$2,751	\$3,126	\$2,732	\$23,521	\$23,521	\$20,381	\$17,143	\$13,800	\$10,255	\$10,255	(\$2,986)	\$47,526	\$211,660
Total assets	\$6,995	\$4,094	\$2,751	\$3,126	\$2,732	\$23,521	\$23,521	\$20,381	\$17,143	\$13,800	\$10,255	\$10,255	(\$2,986)	\$47,526	\$211,660
Current Liabilities															
Accounts payable and accrued interest	256	1,178	1,048	1,296	1,296	1,296	1,296	1,296	1,296	1,296	1,296	1,296	1,296	1,296	1,296
Deferred compensation		306	306	306	306	306	306	306	306	306	306	306	306	306	306
Unreimbursed expenses	38	76	84	83	83	83	83	83	83	83	83	83	83	83	83
Total Current Liabilities	\$294	\$1,560	\$1,439	\$1,686	\$1,686	\$1,686	\$1,686	\$1,686	\$1,686	\$1,686	\$1,686	\$1,686	\$1,686	\$1,686	\$1,686
Total Liabilities	\$294	\$1,560	\$1,439	\$1,686	\$1,686	\$1,686	\$1,686	\$1,686	\$1,686	\$1,686	\$1,686	\$1,686	\$1,686	\$1,686	\$1,686
Stockholders' equity															
Common Stock	13	13	13	13	13	13	13	13	13	13	13	\$13	\$13	\$13	\$13
Additional paid-in capital	57,052	68,675	70,711	74,864	79,275	105,296	105,296	107,480	109,765	112,151	114,836	114,836	124,379	133,925	143,474
Accumulated deficit	(50,365)	(66,153)	(69,412)	(73,437)	(78,242)	(83,474)	(83,474)	(88,798)	(94,321)	(100,049)	(106,279)	(106,279)	(129,063)	(88,098)	66,487
Total Equity	6,700	2,534	1,313	1,440	1,046	21,835	21,835	18,696	15,458	12,115	8,570	8,570	(4,671)	45,841	209,975
Total Liab & Equity	\$6,995	\$4,094	\$2,751	\$3,126	\$2,732	\$23,521	\$23,521	\$20,381	\$17,143	\$13,801	\$10,256	\$10,256	(\$2,985)	\$47,526	\$211,660
Shares Issued (in thousands)	12,848	13,015	13,171	13,371	13,384	15,398	13,831	15,413	15,429	15,444	15,459	15,436	15,498	15,560	15,622
Shares Outstanding (in thousands)	12,848	13,015	13,171	13,371	13,384	15,398	13,831	15,413	15,429	15,444	15,459	15,436	15,498	15,560	15,622

Source: Company reports and Noble Capital Markets estimates

Greenwich LifeSciences, Inc. Cash Flow Statement (in thousands)																			
	2023A	1Q24A	2Q24A	3Q24A	4Q24A	2024A	1Q25A	2Q25A	3Q25E	4Q25E	2025E	1Q26E	2Q26E	3Q26E	4Q26E	2026E	2027E	2028E	2029E
Cash flows from operating activities:																			
Net income (loss)	(8,392)	(2,473)	(5,080)	(7,749)	(15,789)	(15,789)	(3,259)	(7,284)	(12,089)	(17,321)	(17,321)	(5,324)	(10,847)	(16,575)	(22,805)	(22,805)	(22,784)	40,965	154,585
Stock based compensation	2,378	595	1,189	1,784	7,253	7,253	1,544	3,088	5,100	6,700	6,700	2,000	4,100	6,300	8,800	8,800	8,800	8,800	8,800
Amortization	4	1	2	3	4	4	1	2	2	2	2								
Changes in assets and liabilities:																			
Prepaid expenses and other current assets	35	90	1,079	518	921	921	(130)	119	119	119	119								
Accounts payable and accrued liabilities					306	306													
Deferred compensation					38	38													
Unreimbursed expenses (accrued)	(4)	5	(20)	30			8	7	7	7	7								
Net Cash Used in Operating Activities	(6,479)	(1,783)	(2,830)	(5,414)	(7,267)	(7,267)	(1,834)	(4,068)	(6,861)	(10,493)	(10,493)	(3,324)	(6,747)	(10,275)	(14,005)	(14,005)	(13,984)	49,765	163,385
Cash flows from investing activities:																			
Purchase of property and equipment																			
Net cash provided by investing activities	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Cash flows from financing activities:																			
Sale of common stock via ATM program, net of costs		299	566	1,747	1,869	1,869	492	3,101	5,500	5,600	5,600								
Sale of common stock via Private Placement, net of costs			2,500	2,500	2,500	2,500													
Sale of common stock through public offering										24,321	24,321	185	370	555	740	740	743	746	749
Proceeds from exercise of warrants																			
Net cash provided by financing activities	0	299	3,066	4,247	4,369	4,369	492	3,101	5,500	29,921	29,921	185	370	555	740	740	743	746	749
Effect of exchange rate on cash and cash equivalents																			
Net increase (decrease) in cash and cash equivalents	(6,479)	(1,483)	235	(1,167)	(2,897)	(2,897)	(1,342)	(967)	(1,361)	19,428	19,428	(3,139)	(6,377)	(9,720)	(13,265)	(13,265)	(13,241)	50,511	164,134
Cash and equivalents, beginning of period	13,468	6,989	6,989	6,989	6,989	6,989	4,092	4,092	4,092	4,092	4,092	23,520	23,520	23,520	23,520	23,520	10,255	(2,986)	47,526
Cash and equivalents, end of period	6,989	5,506	7,225	5,822	4,092	4,092	2,750	3,125	2,731	23,520	23,520	20,381	17,143	13,800	10,255	10,255	(2,986)	47,526	211,660

Source: Company SEC filings and Noble Capital Markets estimates



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The fundamental assessment rating system is designed to provide insights on the company's fundamentals both on a macro level, which incorporates a company's market opportunity and competitive position, and on a micro/company specific level. The micro/company specific attributes include operating & financial leverage, and corporate governance/management. The number of check marks that a company receives is designed to provide a quick reference and easy determination of the company's fundamentals based upon the following five attributes of the company (weighting reflects the importance of each attribute in the overall scoring of company's fundamental analysis):

Attribute	Weighting
Corporate Governance/Management	20%
Market Opportunity Analysis	20%
Competitive Position	20%
Operating Leverage	20%
Financial Leverage	20%

For each attribute, the analysts score the company from a low of zero to a high of ten based upon the analysis described below. The final rating and resulting check marks is a result of dividing the overall score (out of 100%) by ten.

Rating	Score	Checks
Superior	9.1 to 10	Five Checks
Superior	8.1 to 9	Four & A Half Checks
Above Average	7.1 to 8	Four Checks
Above Average	6.1 to 7	Three & A Half Checks
Average	5.1 to 6	Three Checks
Average	4 to 5	Two & A Half Checks
Below Average	3 to 3.9	Two Checks
Below Average	2 to 2.9	One & A Half Checks
Low Quality	0 to 1.9	One Check

While these are the attributes currently used for the analyst's fundamental analysis, the attributes and weighting may be reviewed, updated with additional attributes, and/or changed in the future based on discussions with the analysts and recommendations from the Director of Research.

Following is the description of each attribute in the fundamental analysis.

Corporate Governance/Management

We believe that a review of corporate governance and assessment of the senior management are important tools to determine investment merit. Good corporate governance aligns management with the interests of stakeholders. As such, analysts are to rank the company on the basis of good corporate governance principles that may include rules and procedures, board composition and staggered term limits, rights and responsibilities, corporate objectives, monitoring of actions and policies, and accountability. In addition, analysts will assess issues with controlling shareholders and whether decisions have been made in the past that were in the interests of all shareholders. In addition, management will be assessed based on industry experience, expertise, and/or track record.

High ranking example: Board and management that is aligned with the interests of shareholders with incentives based on stock price appreciation and with an experienced management team known for exceptional shareholder returns.

Low ranking example: Concentrated ownership without independent directors that do not necessarily align with all shareholders' interests.

The Market Opportunity Analysis

In this review, the analyst assesses the company's macro environment as a measure of understanding the industry. Factors considered include the size and growth potential of the industry under various economic conditions, the emerging demands in the market, technological benefits/disruptions, competition, geographical opportunities, and customer demands/needs, and an assessment of supply and distribution channels. In addition, the analyst will review legal and regulatory trends, as well as potential shifts in consumer or social behavior and natural environment changes.

High rank example: A company in an industry that is growing revenues well above GDP rates (which are on average 2% plus) and/or may have unmet or underserved needs in a rapidly growing market opportunity.

Low rank example: A mature industry that is in secular decline and likely to grow below GDP rates.

Competitive Position

The evaluation of the company's competitive position is another macro environment attribute designed to measure the relevance, market share, position and value proposition, and sustainable differentiations of the company and its products/services within its industry. Ease of entry into the industry and the ability of other well-funded players to potentially enter the market would be determined. As such, the assessment would consider the company's strengths and advantages of its products/services against weaknesses and limitations. This may include the company's current brand awareness, pricing and cost structure, current market strategies and geographic penetration that may affect demand for its products/services. In addition, the company's competitors would be evaluated.

High rank example: An analyst would consider the company's product to be superior to its competitors and that should allow the company to gain market share.

Low rank example: A company with a "me-too" product that does not have any significant technology advantages in an industry that has low barriers to entry.

Operating Leverage

Simplistically, operating leverage is determined by the operating income relative to changes in revenue. The analyst will calculate the impact on sensitivity on gross margins and variable costs to determine operating leverage. The analyst will take into account the ability of the company to cut fixed and variable costs in a challenged revenue environment and technological changes that may impact operating expenses. In addition, the analyst is to assess corporate strategies that include capital investment, which may be required for sustainable revenue growth, marketing expenses, and the company's ability to attract and retain talent and/or employees. The analyst should focus on the revenue opportunity and determine the price elasticity of demand for the company's products or services. In other words, the analyst is to rank the company based on improved operating margins going forward on an absolute and relative basis.

High rank example: A company that has improving margins for the foreseeable future, with significant price elasticity.

Low rank example: A company that is in a challenged revenue environment with a fixed cost structure and limited ability to cut costs, indicating an outlook for declining margins.

Financial Leverage

A strict definition of financial leverage is total debt divided by total shareholder's equity. Financial leverage analysis is to determine the company's ability to improve shareholder value by means of utilizing its balance sheet to grow organically or to acquire assets. Analysts may look at the company's debt to cash flow leverage ratio, interest coverage ratios, or debt to equity ratios. In addition, the interest rate environment and the outlook for interest rates are a factor in determining the company's ability to manage financial leverage. Finally, the analyst is expected to determine the ability to service the debt given the industry and/or company profile, such as cyclical, barriers to entry, history of bankruptcy, consistency in revenue and profit growth, or predictability in sales and profits and large cash reserves. The analyst is expected to take into account capital intensity of the company and the anticipated of capital allocation decisions.

High rank example: A company with predictable and growing revenue and cash flow with modest debt levels. This may indicate that the company could improve shareholder value through growth investments, including acquisitions, using debt financing.

Low rank example: A company in a cyclical industry in a late stage economic cycle that has above average debt leverage and is in an industry that has a history of financial challenges, including bankruptcies.

ANALYST CREDENTIALS, PROFESSIONAL DESIGNATIONS, AND EXPERIENCE

Senior Equity Research Analyst focusing on the Biotechnology and Specialty Pharmaceuticals industry. 16 years of industry experience. BA in Economics from Tulane University and an MBA from Columbia Business School. FINRA licenses 7, 24, 63, 86, 87

CONTINUING COVERAGE

Unless otherwise noted through the dropping of coverage or change in analyst, the analyst who wrote this research report will provide continuing coverage on this company through the publishing of research available through Noble Capital Market's distribution lists, website, third party distribution partners, and through Noble's affiliated website, channelcheck.com.

WARNING

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RESEARCH ANALYST CERTIFICATION**Independence Of View**

All views expressed in this report accurately reflect my personal views about the subject securities or issuers.

Receipt of Compensation

No part of my compensation was, is, or will be directly or indirectly related to any specific recommendations or views expressed in the public appearance and/or research report.

Ownership and Material Conflicts of Interest

Neither I nor anybody in my household has a financial interest in the securities of the subject company or any other company mentioned in this report.

NOBLE RATINGS DEFINITIONS	% OF SECURITIES COVERED	% IB CLIENTS
Outperform: potential return is >15% above the current price	86%	14%
Market Perform: potential return is -15% to 15% of the current price	14%	5%
Underperform: potential return is >15% below the current price	0%	0%

NOTE: On August 20, 2018, Noble Capital Markets, Inc. changed the terminology of its ratings (as shown above) from "Buy" to "Outperform", from "Hold" to "Market Perform" and from "Sell" to "Underperform." The percentage relationships, as compared to current price (definitions), have remained the same.

Additional information is available upon request. The recipient of this report who wishes further information regarding the subject company or the disclosure information mentioned herein, should contact by mail or phone.

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